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Bearing fault diagnostic framework under unknown working conditions based on condition-guided diffusion model

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ABSTRACT

Fault diagnosis is of vital importance in ensuring the safety of smart manufacturing. Current diagnostic methodologies require data spanning various working conditions. However, industrial settings offer scarce bearing failure data, leading to failure or degradation of performance of traditional intelligent diagnostic methods. Obtaining a complete sample of industrial environments is intensive and unrealistic. Acquiring a comprehensive sample of industrial environments is both resource-intensive and impractical. To handle this situation, an unknown condition diagnosis framework based on diffusion model (DiffUCD) is proposed, effectively integrating the generation capability of the diffusion model and learning from the condition-guided information (CGI), Specifically, signals under limited working conditions are gradually convert to noise through a forward noising process. Then, CGDiffusion reconstructs signals from the noise by a reverse denoising process. In addition, a condition-guided embedding UNet (CGE-UNet) structure is designed to extract CGI for noise level prediction during the reverse process. Moreover, an Unsupervised Clustering Filter (UCFilter) is constructed to select the qualified signals after generation. Signals under unknown working condition can be generated with specialized CGI. Ultimately, extensive experiments and validations on two public bearing datasets (SDUST and PU) are carried out, which validate the effectiveness of our method compared with the state-of-the-art baselines and the hyperparameter analysis confirms the advances of DiffUCD.

1. Introduction

Rotating machines are of vital importance in today's industrial fields, highlighting the need to ensure their safety and reliability through competent condition monitoring and fault diagnosis techniques [1]. Within the diverse spectrum of data-driven fault diagnosis methodologies, deep learning based strategies have attracted considerable attention and focus in the field of machine health monitoring[2,3].

Nevertheless, the effectiveness of deep learning is dependent upon the assumption of consistent data distribution between the training and testing sets. The practical applications of deep learning in industry may be constrained by a number of challenges, which could limit the utility of these methods. (1) Rotating components frequently operate under diverse conditions, including fluctuations in rotational speed and load [4]. (2) Obtaining sufficient labelled data with precise health information across all operating conditions can be impractical. (3) Domain shift issues arise when attempting to compensate for information gaps by utilizing labelled data from multiple machines or different working conditions[5]. Hence, while deep learning holds promise for fault diagnosis in rotating machinery, addressing the challenges of varying operating conditions and data distribution discrepancies is crucial for realizing its full potential in practical industrial settings[6].

In recent years, several state-of-the-art methods have been developed to address the challenges outlined above. In light of the existence of domain shift, the empirical risk minimization (ERM) [7]principle is rendered invalid. This is because a diagnosis model trained in a specific domain is unable to generalize the diagnosis knowledge to an unseen working condition. Thus, catastrophic degradation of diagnostic performance occurs. Other such approach is domain adaptation (DA) and domain generalization (DG), which aims to mitigate the distribution discrepancy between the source and target domains during model training[8,9]. DA enables the transfer of knowledge learned from large source datasets to build diagnostic models for smaller target datasets with similar characteristics. Wang, et al. [10] proposed the use of intra-class maximum mean discrepancy (MMD) in conjunction with multi-scale ResNet architectures to reduce the conditional distribution

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Table 1
Feasibility for cross-domain diagnosis of different methods.

Method	domain shift	multiple domains	unseen faults & unseen domains	domain augmentation
ERM	×	×	×	×
DA	√	×	×	×
DG	√	√	×	×
GAN	√	√	×	√
DDPM	√	√	√	√

discrepancy of vibration signals. Hu, et al.[11] introduced tensor-aligned invariant subspace learning, which enables the discovery of a shared tensor representation for cross-domain diagnosis tasks. Inspired by adversarial and generalized learning principles, Zhuo et al.[12] developed a domain generative approach with the incorporation of auxiliary information for the purpose of diagnosing faults that had not been previously encountered. Li et al.[13] conducted a comparative analysis of five distinct data augmentations for the purpose of diagnosis, wherein signal translation was observed to yield the most notable improvements.[14] Advancements in domain adaptation techniques show promise for improving the robustness and generalization capabilities of fault diagnosis models. However, the aforementioned domain adaptation methods are limited by the closed-set assumption [15]. In essence, there is an irreconcilable discrepancy in the feature distributions between the source domain data and the target domain [16].

Under this premise, it is necessary to develop a technique that takes the out-of-distributed (OOD) fault classification into account[17]. Up to now, Generative Adversarial Networks (GANs) adopts an unsupervised learning method and automatically learns from the source data, so that it can produce amazing effects without artificial labelled data. In the applications of PHM, conditional GANs have been used to control the generation process to generate desired distinct classes[18]. Wang, et al. [19] introduced an enhanced version of Least-Square Generative Adversarial Networks (LSGANs) that mitigates the mode collapse issue and preserves more details in augmented crack signals. The improved LSGANs notably retain more crack signal details compared to traditional methods and exhibit significant robustness. Furthermore, these methods are only suitable for generating data from previously observed conditions and not for generating previously unseen conditions in a specific domain. The latter is the focus of our research [20,21].

For the issue of the diagnostic works under unseen working conditions, Rombach, et al. [22]propose a new framework for Open-Partial DA based on generating distinct fault signatures with a Wasserstein GAN, which enables a better transferability between two different domains, where limited data or classes are in the new domains. However, the main drawback of GANs is that they are unstable during the training process. Also, GAN models process rare domain augmentation abilities due to the limited understanding of unseen domain knowledge. The denoising diffusion probabilistic model (DDPM) does not suffer from GANs-like problems of training non-convergence and pattern collapse [23]. The latest research findings suggest that existing data from a range of available domains may act as a catalyst for the generation of augmented data in a novel domain. Due to its simple objective function, clever forward–backward process design and powerful backbone network, it performs well on various tasks. Furthermore, diffusion model has few-shot and zero-shot capability and also has stable generation results on poor quality data containing noise[24,25].

Hence, for PHM applications, there multiple working conditions and limited fault data to acquire which is suitable for DDPM method to perform unseen domain diagnosis function. The feasibility for cross-domain diagnosis of different models is compared in Table 1. To achieve the stable and effective diagnostic framework for unseen working conditions based on feature embedding, we first propose a diffusion-based unknown condition diagnosis Framework (DiffUCD), which can generate a complete bearing sample of industrial environments and maintain the real-world working conditions. DiffUCD effectively integrates the generation capability of the diffusion model and learning from the condition-guided information (CGI). Essentially, the forged signals generated by DiffUCD guarantee the generation accuracy while retaining the utility. To summarize, the main contributions of this work are concluded as follows:

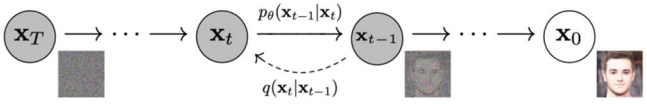


Fig. 1. Two parts in Denoising Diffusion Probabilistic Models.

1) A denoising diffusion-based generative model DiffUCD is proposed, which can generate high-quality signal data through the diffusion process and an unsupervised clustering signal selection for more quality signals. After training, the framework can take random Gaussian noise as input and progressively eliminate this noise through a reverse process and generate signals in unseen working conditions.
2) The CGE-UNet is proposed, which replaces the traditional U-Net structure in diffusion model better with the attention mechanism and the condition encoding methods for unseen condition information embedding. An unsupervised clustering generated signal selection method called UCFilter is constructed based on K-means clustering to select the valuable signals after generation.
3) Extensive fault diagnosis experiments are conducted on the widely utilized rolling bearing dataset of Paderborn university (PU) and SDUST bearing dataset. Our proposed method is thoroughly compared with the state-of-the-art generative models in the field of fault diagnosis.

The organization of the remainder of this paper is outlined as follows: Section II provides an introduction to pertinent techniques and background knowledge. Section III delves into the structure and specifics of the proposed diagnostic framework for unseen conditions. Section IV presents the experimental validation and real-world application in detail. Finally, Section V brings the paper to a conclusion.

2. Preliminaries

2.1. Diffusion model

The diffusion model [26] is a novel method inspired by the diffusion of gases from high to low concentrations. The diffusion model comprises two primary components, as shown in Fig. 1. The first part is the forward process, also known as the diffusion process, which gradually blends noise into an original sample with the intention of corrupting it.

Over the course of continuous noise addition, the previously intact sample data are transformed into a fully noisy and indistinguishable sample. The mathematical description of this process is as follows:

$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1-\overline{\alpha}_t}\epsilon\# \tag{1}$$

where x_t represents the data obtained from t steps of the forward process, while x_0 denotes the original data. $\overline{\alpha}_t$ is a hyperparameter that is computed using $\overline{\alpha}_t = \prod_{i=1}^{n} \alpha_i$. Typically, α_i is a small constant value. ϵ is the Gaussian-distributed noise.

The second part is the inverse process, i.e. the denoising process, in which the model must predict and infer the noise introduced during the forward process. By estimating the added noise, the trained and converged diffusion model can gradually remove noise from a randomly sampled signals that conforms to a Gaussian distribution and generate pseudo samples that conform to the original data distribution. The mathematical expression for the inverse process is given below:

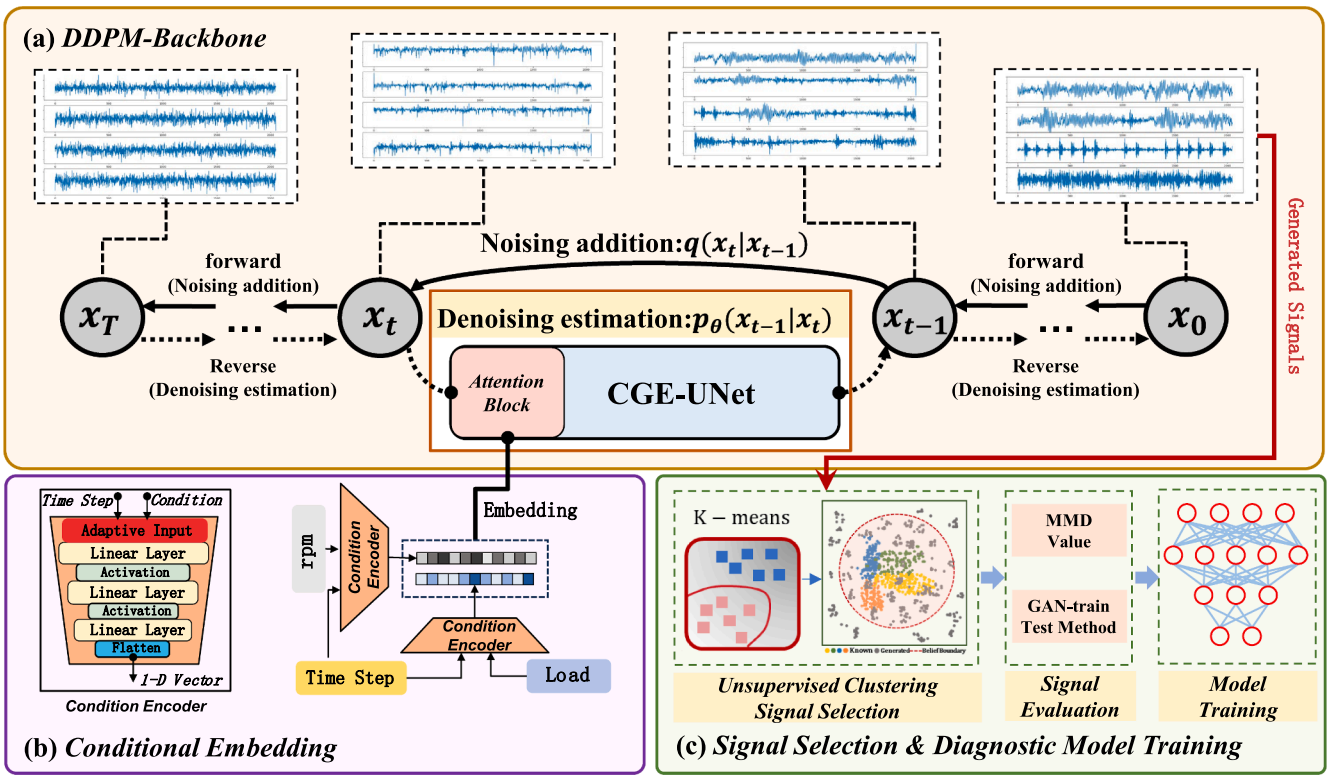


Fig. 2. Overall Structure Schematic of DiffUCD Framework.

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z\# \tag{2}$$

where x_{t-1} is the denoised sample obtained at step $t-1$, x_t is the data obtained at step t, $\epsilon_\theta(x_t, t)$ represents the noise estimated by U-Net, α_t is a known constant parameter, z denotes standard Gaussian noise, and σ_t is the predefined variance at step t, which can be calculated as follows:

$$\sigma_t^2 = \widetilde{\beta}_t = \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t}\beta_t\# \tag{3}$$

where β_t is computed using $1-\alpha_t$. The diffusion model employs a multistep denoising process to gradually remove noise from a noisy sample, thereby obtaining samples that conform to the original distribution.

The noise prediction network, which is based on U-Net, is the essential element of the entire diffusion model that requires training. U-Net is a neural network architecture that incorporates an encoder, middle layers, and a decoder. The design of U-Net is such that it preserves both local and global information by incorporating skip connections between the corresponding layers of the encoder and decoder. The

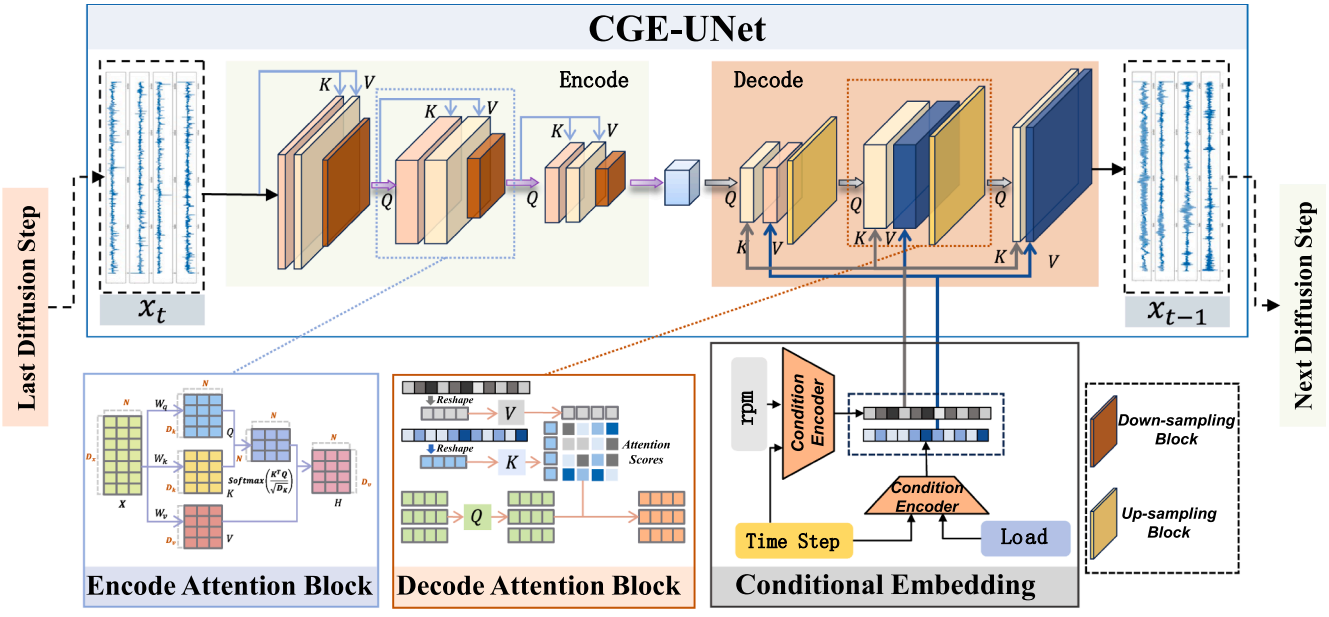


Fig. 3. The details of proposed CGE-UNet based on attention block.

loss function is as follows:

$$\begin{aligned} Loss &= \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \\ &= \|\epsilon - \epsilon_\theta\left(\sqrt{\overline{\alpha}_t}\, x_{t-1} + \sqrt{1-\overline{\alpha}_t}\,\epsilon, t\right)\|^2 \# \end{aligned} \tag{4}$$

where ϵ represents the noise that needs to be estimated, and $\epsilon_\theta(x_t, t)$ represents the noise predicted by U-Net at step t. x_t is the sample obtained after adding noise at step t during the forward process.

The diffusion model's loss function is structured in a way that makes it clear that the model's main objective is to predict the noise blended at each step. By reducing the discrepancy between the expected and actual added noise, the diffusion model is capable of recreating the original data from noise.

3. The proposed method

In the problem of domain generalization, traditional approaches focus on acquiring data features or augmenting unseen or out-of-distribution data. Our proposed methods utilize the generation and the feature obtaining abilities of diffusion model to generate data that are out of domain. This section outlines the proposed DiffUCD framework, as depicted in Fig. 2, which mainly consists of three parts: (1) the DDPM-Backbone, which performs the main noise and denoising data generation processes of the diffusion model. (2) The CEG-UNet, which replaces the traditional U-Net structure in the diffusion model with the embedding function of a conditional encoder based on an attention block. (3) UCFiler, which integrated an unsupervised clustering and selection module for obtaining higher quality signals.

3.1. The CEG-UNet in DDPM backbone

The detailed processes of DDPM are illustrated in II.A, which is the fundamental of the proposed DiffUCD. The forward process can be briefly parameterized as follows:

$$q(x_1, \cdots, x_N|x_0) = \prod_{t=1}^{N} q(x_n|x_{n-1})\# \tag{5}$$

where $q(x_n|x_{n-1})$ denotes the conditional Gaussian distribution and the forward-process variances β_n adjust the noise level. The reverse process can be briefly parameterized as:

$$p_\theta(x_0, \cdots, x_{n-1}|x_N) = p(x_N)\prod_{t=1}^{N} p_\theta(x_{n-1}|x_n)\# \tag{6}$$

where $x_N \sim \mathcal{N}(0,1)$ and $p_\theta(x_{n-1}|x_n)$ is assumed to follow normal-distributed with learnable parameters.

The CEG-UNet utilizes an attention based embedding block, which can transform the encoded conditional information to the guidance weight for the generative process. The specifics of CEG-UNet are shown in Fig. 3.

Dataflows in CEG-UNet contain an encode process and a decode process. In the encode process, a self-attention method is utilized to extract and re-weight the features of generated signals from last diffusion step. Given the signal features ψ_{signal}, the learnable weight matrices W_Q, W_K, and W_V are applied to the query Q_E, key K_E, and value V_E matrices as:

$$\{\mathbf{Q_E}, \mathbf{K_E}, \mathbf{V_E}\} = \{\psi_{\text{signal}}\ W_Q, \psi_{\text{signal}}\ W_K, \psi_{\text{signal}}\ W_V\}\# \tag{7}$$

Following the encode process, we also introduce a cross-attention-based decode process to further integrate the conditional information which need to be embedded. In this process, the encoded signal features act as the queries, while the embedded conditional features ψ_{condi}. In the conditional embedding methods, as depicted in Fig. 2 (b), the input of two encoders are time steps and rpm or load information. The output of these two encoders is regarded as the conditional features ψ_{condi} and intergraded with the diffusion process by times W_K and W_V serving as

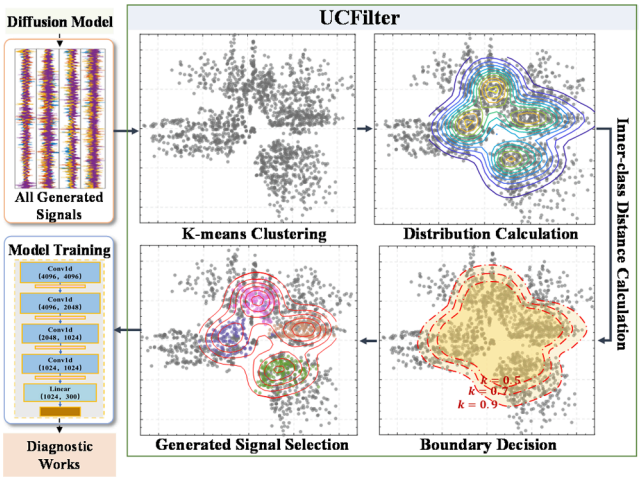


Fig. 4. The flowchart of sample selection base on ucfilter.

the keys and values as follows:

$$\{\mathbf{Q_D}, \mathbf{K_D}, \mathbf{V_D}\} = \{\psi_{\text{signal}}\ W_Q, \psi_{condi} W_K, \psi_{condi}\ W_V\}\# \tag{9}$$

Attention weights are then computed using the dot product of queries and keys, normalized by the SoftMax function. These weights are then multiplied by the values to produce the fused feature representation. The simple formulaic expression of this procedure is as fellow:

$$Attention(Q, K, V) = softmax\left(\frac{QK^\top}{\sqrt{d_k}}\right)\cdot V\# \tag{10}$$

where d_k is the dimension of the key vectors. In practice, the self-attention mechanism is expanded into a multi-head self-attention mechanism, which allows the attention mechanism to consider diverse attention distributions and enables the model to capture information from multiple viewpoints. This mechanism captures global interactions within the signal domain.

This approach permits the model to weigh the signal features in accordance with conditional semantics, thereby ensuring that the fused representation is influenced by the embedded conditions. In order to capture diverse perspectives and guarantee a comprehensive fusion of features, we employ a multi-head mechanism in the cross-attention unit. This multi-head cross-attention mechanism ensures that the fused representation captures a broad spectrum of interactions between sample and textual features, resulting in a robust and semantically rich feature representation that is suitable for downstream detection tasks.

3.2. Conditional embedding method

The essence of generating fault samples of unseen working conditions in our proposed framework is the conditional encoding. As shown in Fig. 3, we utilize the rotation speed and operation load as the variable working conditions.

Different rpms and loads are separately encoded with the time through an auto-encoder, which is called condition encoder (CE). Apart from the time step which comes from the DDPM backbone, the input of CE is the rpm and load value under the standard scale. The output of CE is conditional features ψ_{condi} that is to be embedded through the encode/decode attention block. The network structure of is as depicted in Fig. 3, which consists of an adaptive input layer and three linear layers with activation functions. The adaptive input layer ensures that CE can change its input channels adaptively according to the vector length of conditional input. The structure of CE can be expressed as follow:

$$x_{adain} = AdaIn[Concat\{x_{condi}, t\}]\# \tag{11}$$

$$\psi_{condi} = \sigma\left(\sum_{i=1}^{L} W_i \times x_{adain} + b_i\right)\# \tag{12}$$

where the W_i and L are the weight and length of linear layer, b_i is the bias of linear. $Concat\{x_{condi}, t\}$ means the conditional vector x_{condi} and time step t are concatenated together. $AdaIn[\bullet]$ represents the adaptive input layer, and $\sigma(\bullet)$ is the utilized activation function in our CE structure.

3.3. Unsupervised clustering Filter

Once the DDPM is completed in the DiffUCD training, the working conditions of the signal generation are controlled by the CGE-UNet. As shown in Fig. 4, the generated signal is clustered by the K-means algorithm, which is an unsupervised clustering method. The top of n samples closest to the signal sample centre are selected as valuable signals.

The distances between inter-class samples are measured by distribution probability based on Kullback-Leibler divergence between the joint probabilities R_{ij} in the high-dimensional space and the joint probabilities T_{ij} in the low-dimensional space. The values of R_{ij} are defined to be the symmetrized conditional probabilities, whereas the values of T_{ij} are obtained by means of the Student's t-distribution with one degree of freedom. The calculation is summarized as follows:

$$r_{ij} = \frac{r_{i|j} + r_{j|i}}{2n}\# \tag{13}$$

$$t_{ij} = \frac{\left(1 + |y_i - y_j|^2\right)^{-1}}{\sum_{k \neq l}\left(1 + |y_k - y_l|^2\right)^{-1}}\# \tag{14}$$

where $r_{i|j}$ is the distribution probability of sample point j when the sample point i is given. y is the generated samples. The values of r_{ii} and t_{ii} are set to zero. The calculation of the Kullback–Leibler divergence C_d between the two joint probability distributions R and T is given as follows:

$$C_d = KL(R|T) = \sum_i \sum_j r_{ij} log r_{ij} - r_{ij} log t_{ij}\# \tag{15}$$

After calculated the Kullback–Leibler divergence C_d of every sample in generated signals, the selection boundaries are decided according to the expectation numbers of acceptable samples n, and k value of the Boundary Decision indicates the portion of selected samples ($k = n/N$).

3.4. Quality evaluation of generated samples

In order to comprehensively evaluate the quality of the generated samples under conditions that have not been previously encountered, a sample augmentation evaluation method has been constructed. This method assesses the capacity of the proposed model to comprehend genuine distributions, the diversity and verisimilitude of the generated samples, and the diagnostic precision achieved under disparate equilibrium conditions.

In order to evaluate the effectiveness of the sample augmentation model in aligning with the real data distribution, this study employs the Maximum Mean Difference (MMD) metric to quantify the discrepancy between the model's learned distribution and the original data set. MMD is a statistical metric that employs kernels to quantify the discrepancy between two specified distributions. A lower MMD value indicates a greater degree of similarity between the two distributions. The utilization of the MMD metric enables the visualization of the model's capacity to effectively fit the actual data distribution. The mathematical description is provided below:

$$L^2_{\mathrm{MMD}} = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{i'=1}^{n} k(x_i, x_{i'}) - \frac{2}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} k\left(x_i, y_j\right) + \frac{1}{m^2}\sum_{j=1}^{m}\sum_{j'=1}^{m} k\left(y_j, y_{j'}\right) \tag{16}$$

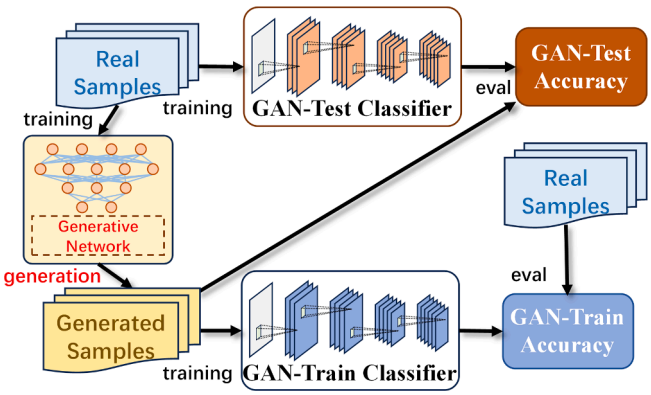


Fig. 5. The illustration of GAN-train/GAN-test methodology.

where, $k(x, y)$ represents the result obtained with samples under the kernel function mapping, while m and n represent the numbers of real and generated samples, respectively. In order to minimize the randomness of spatial mapping, we employ multiple distinct Gaussian kernel functions for MMD metric calculation. The final MMD[27] score is then obtained by taking the average of the obtained values.

Furthermore, the GAN-train/GAN-test methodology is introduced to assess the feature learning proficiency and sample diversity of generative models. In GAN-train, the synthetic data trains a classification network, while real data tests it. Limited diversity in the generated samples constrains the classifier's generalization, impacting classification performance on real-test samples. Conversely, GAN-test trains a classifier on real data and tests it on synthetic data. Low authenticity of generated samples hinders performance. The flow chart of GAN-train/GAN-test methodology is shown in Fig. 5. This dual approach quantifies model diversity and distribution fitting, offering a holistic evaluation.

4. Experiment

In this section, to assess the effectiveness of the general framework, extensive fault diagnosis experiments are conducted using three bearing fault datasets, including two public bearing fault dataset which are SDUST[28] and the Paderborn University bearing dataset (PU dataset) [29]. In addressing the issue of domain generalization, traditional methodologies have focused on the extraction of salient data features and the augmentation of datasets with examples that are either unseen or outside the distribution of the data. Specific working conditions as the unseen target domain within these datasets are designated, serving as the benchmark for evaluating the model's performance. Meanwhile, the data pertaining to the remaining as seen conditions are utilized for model training.

To validate the actual performance of the generative DDPM model, this paper presents the following four generative algorithms for comparison: a vanilla GAN (VGAN)[30], a deep convolutional generative adversarial network (DCGAN)[31], WGAN-GP[32], and an adversarial autoencoder (AAE)[33]. All these GANs are trained on the same dataset as the DDPM model. The number of epochs for these networks is set to 1000. The DDPM and each GAN model are then used to generate new samples called after their training processes have been completed. The Learning Rate Schedule is set as decrease by a factor of 0.1 every 20 epochs. The Adam optimizer is used in the neural network optimization process.

4.1. Case1: SDUST dataset

1) Dataset Description

The SDUST dataset comprises a motor, a shaft coupling, a rotor, a

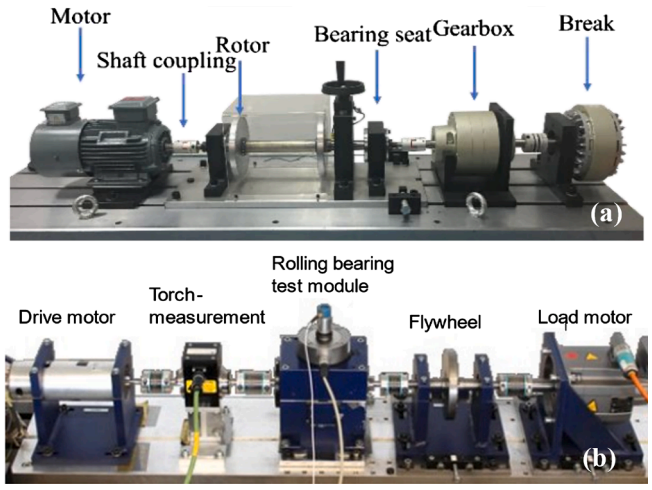
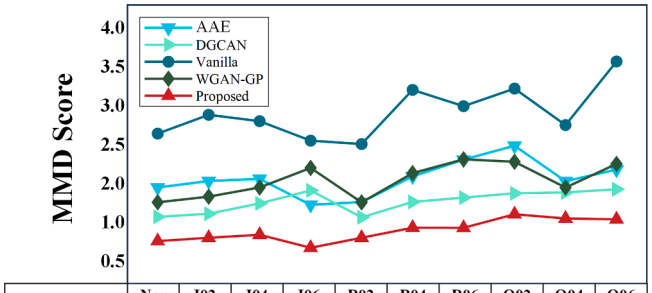


Fig. 6. Experimental platform of SDUST dataset (a) and PU dataset (b).

Table 2
Case1: Diagnostic cases of SDUST dataset.

Diagnostic cases	Seen Domain	Unseen Domain
T1	1500r/min,2000r/min,2500r/min	1000r/min
T2	1000r/min,2000r/min,2500r/min	1500r/min
T3	1000r/min,1500r/min,2500r/min	2000r/min
T4	1000r/min,1500r/min,2000r/min	2500r/min



	Nor	I02	I04	I06	B02	B04	B06	O02	O04	O06
AAE	1.539	1.845	1.828	1.501	1.372	1.662	1.774	2.300	1.821	1.816
DCGAN	0.992	1.096	1.199	1.437	1.104	1.372	1.409	1.662	1.515	1.763
Vanilla	2.513	3.073	2.401	2.426	2.109	3.245	3.001	2.703	2.357	3.231
WGAN-GP	1.423	1.591	1.510	2.043	1.425	1.537	2.079	1.912	1.779	1.655
Proposed	0.749	0.791	0.845	0.708	0.766	1.046	0.954	1.008	0.998	1.166

Fig. 7. MMD scores between the real and generated samples in Case 1.

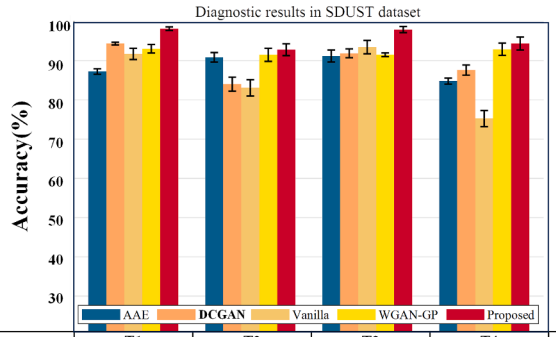
testing bearing, a gearbox, and a brake, as shown in Fig. 6. Specifically, the bearing type used is N205EU, and data was gathered under four distinct health conditions: normal (NOR), inner ring fault (I), rolling element fault (B), and outer ring fault (O). SDUST bearing dataset contains only one variable working condition, i.e. operating speed. Four variable working conditions were investigated at speeds of 1000, 1500, 2000 and 2500 rpm. Under these four operating conditions, there are 10 different types of health categories (abbreviated as Nor, I02, I04, I06, B02, B04, B06, O02, O04, and O06, respectively). The data in each health category have 100 samples. Each sample has 2048 data points, fully covering the error information.

2) Data Preprocessing.

The experimental setup includes four diagnostic scenarios for unseen working conditions across domains, namely T1000, T1500, T2000, and T2500, are shown in Table 2. For example, the T1500 task entails training the model on observed domains at varying speeds of 1000, 2000, and 2500 r/min, with the objective of generalizing the acquired knowledge to an unobserved target domain operating at a speed of 1500 r/min. Similarly, the validation dataset is constructed by all working conditions to perform model evaluation, and the data in the validation dataset have no overlap with the training set.

Table 3
The GAN-Train and GAN-Test Accuracy in Case 1.

Models	GAN-Train Accuracy (%)	GAN-Test Accuracy (%)
Raw data	98.56 ± 0.24	/
AAE	92.44 ± 0.57	62.35 ± 0.42
DCGAN	85.74 ± 0.25	79.74 ± 0.58
VGAN	74.18 ± 0.15	67.25 ± 0.45
WGAN-GP	93.58 ± 0.38	93.10 ± 0.84
Proposed	96.57 ± 0.42	96.41 ± 0.37



	T1	T2	T3	T4	Average
AEE	87.23±0.67	90.86±1.17	91.19±0.42	84.80±1.10	88.5234
DCGAN	94.35±0.72	84.21±0.98	91.87±1.07	87.59±0.56	89.4583
Vanilla	91.70±1.80	83.06±1.03	93.46±1.26	75.22±0.55	85.8640
WGAN-GP	93.04±1.19	91.48±0.63	91.51±0.91	92.91±1.22	92.2399
Proposed	98.14±0.24	92.79±0.81	97.92±0.19	94.38±0.97	95.8150

Fig. 8. Comparison of diagnostic accuracy of different models for the diagnostic task in Case 1.

3) Quality Evaluation

The MMD scores between the sample generated by the DDPM model and the other GAN models used for comparison, as well as the unused original samples under unseen working conditions, are illustrated in Fig. 3 for a total of ten fault types.

As illustrated in Fig. 7, the minimum mean discrepancy (MMD) scores of each fault type between the newly generated samples by each model and the original samples are presented. The MMD scores between the samples generated by the proposed framework and those original samples not used for training are the lowest among all fault classes, averaging only 0.947, which is considerably lower than the scores obtained by other models, leveraging the generative might and feature extraction expertise of diffusion models to craft data that transcends the confines of existing domains, ultimately bolstering generalization capabilities.

The performance of the proposed framework and other generative models used for comparison is shown in Table 3. in terms of GAN train/GAN test accuracies. It can be seen that our method has the highest GAN-test accuracy in average (96.57 %) over other models, i.e. the generated unseen samples can be distinguished more accurately when only seen samples are used as the training set. This indicates that the proposed model has the highest ability to capture and embed features. Furthermore, the GAN-train performance of the proposed approach is also much higher than that of the other models, even approaching the level of the original data. The higher GAN-train scores illustrate that the original samples can be well distinguished when only the generated samples are used as the training set, indicating that our method has a notable advantage in terms of diversity and authenticity during sample generation.

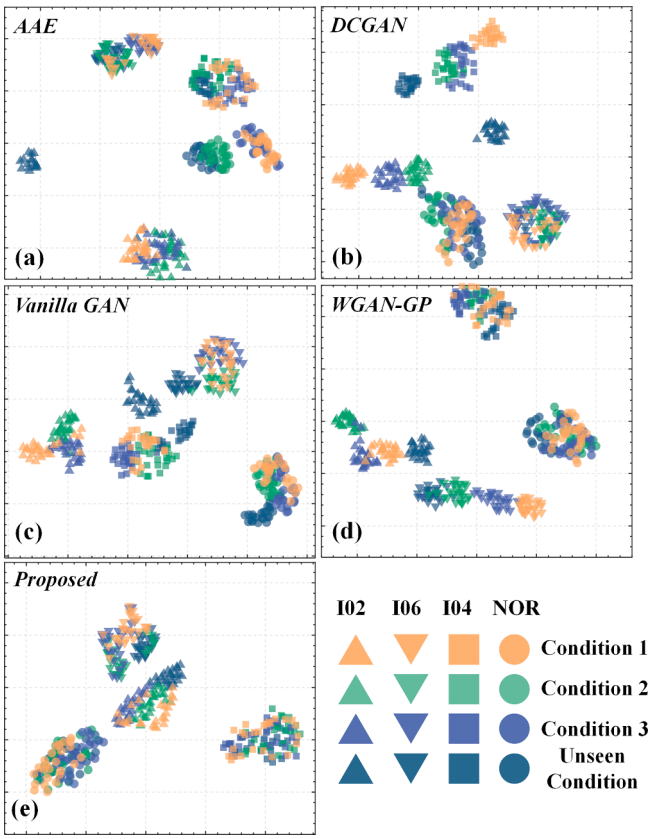


Fig. 9. Comparison of diagnostic accuracy of different models for the diagnostic task in Case 1.

As each diagnostic tasks in Case 1, set in Table 2, The classification accuracy under a newly trained diagnostic model is shown in several conclusions can be drawn. Firstly, the AAE method achieves notably lower average test accuracies of 88.52 %, indicating interference among data distributions during training, affecting model generalization. Secondly, GAN methods as DCGAN, Vanilla GAN and WGAN-GP exhibit improvements over other model, with average accuracy of 89.46 %, 85.86 %, and 92.24 % in Case 1, These methods aim to eliminate distributional discrepancies between source domains and learn domain-invariant representations. Finally, the proposed method achieves the best diagnostic performance in almost all diagnostic tasks, with the highest average accuracy of 95.86 %. Furthermore, it demonstrates superior stability compared to other methods in most cases. Fig. 8. illustrates the classification accuracy of different diagnostic tasks for Case 1, aiding in the comparison of diagnostic results.

Compared methods based on GAN or Auto Encoder models aim to learn domain-invariant features, while our diagnosis seeks to generate feature-embedded signals to cover unseen distributions. For further verify the effectiveness of the proposed model, t-distributed stochastic neighbor embedding (t-SNE)[34] technology is used to visualize the output of the hidden layer of the classifier. However, as illustrated in Fig. 9, the features learned by methods AAE, DCGAN and Vanilla GAN fail to capture the generalized representation of the I02 fault due to significant domain discrepancies between seen and unseen working conditions. Although method WGAN-GP may learn more robust features, it struggles to cluster effectively in the unseen condition of I02 for the unseen samples of I02 still process inner-fault boundaries comparing to the t-SNE plot of proposed methods.

In conclusion, the proposed method achieves the highest GAN-train and GAN-test scores on the SDUST dataset, which shows that it is effective in generating high-quality and diversified fault samples. Meanwhile, the method also achieves the best performance in the MMD metric, which confirms that it is able to learn the distribution of the original defect samples well. From the feature distribution of the t-SNE visualization results, the samples generated by the proposed method have the best divisibility, which confirms the high quality of the generated samples.

Table 4
Specific MMD scores comparation in Case 2.

Fault Category	AAE	DCGAN	Vanilla GAN	WGAN-GP	Proposed
N	2.639	1.230	1.379	2.629	0.855
IF1	3.004	2.044	2.056	1.248	0.884
IF2	2.412	1.245	1.737	1.299	0.769
IF3	1.988	2.050	1.375	2.088	0.741
OF1	2.487	1.265	1.161	1.511	0.911
OF2	3.059	1.496	2.253	1.346	1.030
CF	2.965	1.492	1.533	1.190	1.001

Table 5
Fault categories in PU bearing dataset.

Label	Bearing Code	Damage	Combination	Damage Level
N	K004	/	/	/
IF1	KI21	fatigue: pitting	single damage	1
IF2	KI18	fatigue: pitting	single damage	2
IF3	KI16	fatigue: pitting	single damage	3
OF1	KA04	fatigue: pitting	single damage	1
OF2	KA16	fatigue: pitting	repetitive damage	2
CF	KB23	fatigue: pitting	multiple damage	2

Table 6
Case2: Diagnostic tasks of PU dataset.

Condition	Working Conditions (rpm, loads(N))	Diagnostic Tasks	Seen Conditions	Unseen Conditions
p1	1500, 1000	***R1***	*p2, p3, p4*	*p1*
p2	900, 400	***R2***	*p1, p3, p4*	*p2*
p3	900, 1000	***R3***	*p1, p2, p4*	*p3*
p4	1500, 400	***R4***	*p1, p2, p3*	*p4*

4.2. Case2: PU dataset

4) Dataset Description

In case 2, we utilized the PU dataset which is offered by Paderborn University. As shown in Fig. 6 the test rig mainly consists of a motor, a torque measurement shaft, a bearing test module, a flywheel and a load motor. There are seven health conditions, normal (N), inner race failure (IF) with three damage levels (IF1, IF2 and IF3), outer race failure (OF) with two damage levels (OF1 and OF2) and compound failure (CF) including IF and OF, and their details are shown in Table 5.

5) Data Preprocessing.

The sampling frequency is 64 kHz. According to four working conditions, four domains can be made as P1, P2, P3 and P4. Based on this, four diagnostic tasks are constructed as shown in Table 6, each condition category in every task contains 2000 samples. Faulty bearings with real damage were acquired from an accelerated lifetime test. The vibration data were collected under four different working conditions in terms of rotational frequency (Hz), load torque (Nm) and radial force (N). The sampling frequency is 64 kHz. According to four working conditions, four domains can be made as P1, P2, P3 and P4. Based on this, four diagnostic tasks are constructed as shown in Each condition category in each domain contains 2000 samples.

The optimization process is implemented based on the training set, where the model with the maximum accuracy on the validation set is considered as the optimal one for the current task. For a DG task, this process is repeated ten times by randomly selecting 70 % and 30 % of the

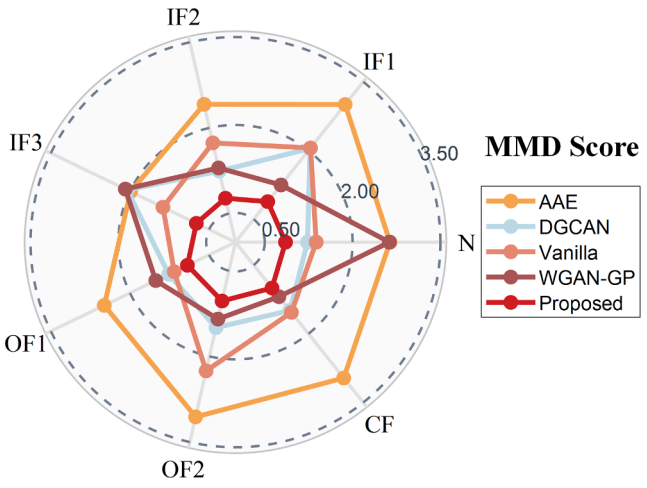
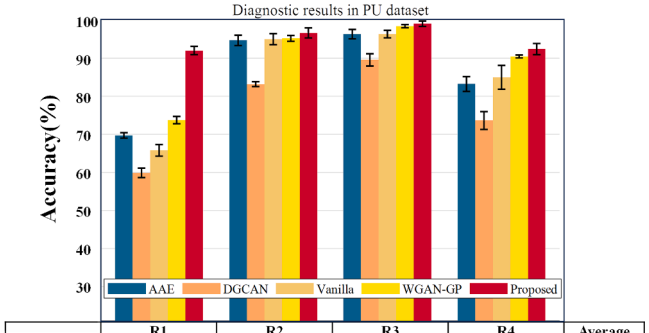


Fig. 10. MMD scores between the real and generated samples in Case 2.

Table 7
The GAN-Train and GAN-Test Accuracy in Case 2.

Models	GAN-Train Accuracy (%)	GAN-Test Accuracy (%)
Raw data	97.64 ± 0.23	/
AAE	92.45 ± 0.68	62.38 ± 0.57
DCGAN	74.23 ± 0.13	67.21 ± 0.29
VGAN	85.62 ± 0.96	79.77 ± 0.44
WGAN-GP	93.65 ± 0.06	93.11 ± 0.95
Proposed	96.58 ± 0.19	96.32 ± 0.09



	R1	R2	R3	R4	Average
AEE	69.71±1.17	94.66±1.11	96.30±1.32	83.20±2.55	85.9704
DCGAN	59.88±1.74	83.17±1.46	89.53±1.51	73.61±3.28	76.5525
Vanilla	65.78±1.50	94.97±1.06	96.31±1.07	84.95±5.23	85.5042
WGAN-GP	73.73±1.03	95.18±0.83	98.39±0.35	90.43±0.61	89.4353
Proposed	92.04±0.87	96.62±1.38	99.03±0.44	92.37±1.52	95.0076

Fig. 11. Generated signals comparison of time-domain on Set R1 of PU dataset and real signals (a)Inner ring fault (b) Outer ring fault (c) Cage fault.

data for the training and validation sets, respectively.

6) Quality Evaluation

The MMD scores between the samples generated by the proposed framework and the samples generated by other GAN or autoencoder based models, as well as the unused original samples, as shown in Fig. 10. The specific values of each fault categories (Table 5) are listed in Table 4. The MMD scores of the samples generated by the proposed framework average 0.884 across all fault categories, which is lower than the 1.616, 1.642, 1.546 and 2.651 of the other models compared. Therefore, it can be concluded that the proposed framework can better approximate the distribution of the original data.

Table 7 illustrates the performance of the proposed model and other GAN or Auto Encoder variants, as indicated by GAN-train/GAN-test

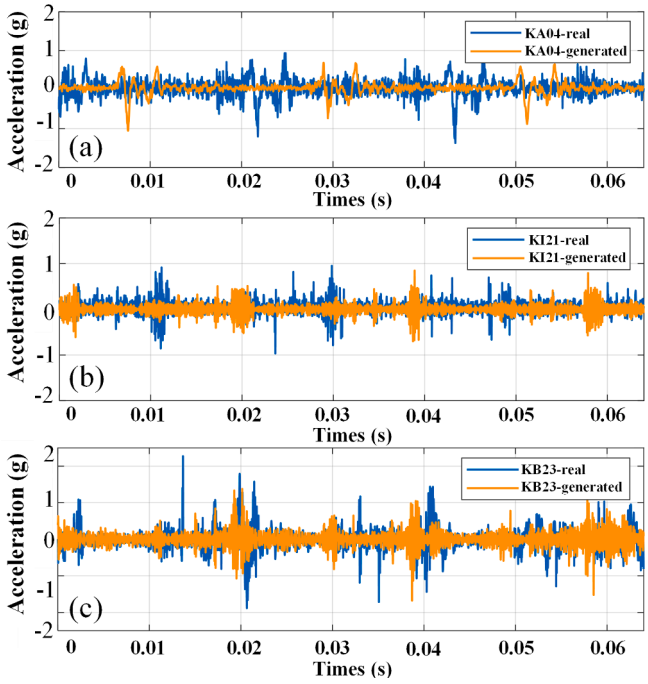


Fig. 12. Comparison of diagnostic accuracy of different models for.

scores. It is evident that the proposed model exhibits a superior performance in GAN-test scores when compared to the other models, which means that the model has enhanced its capacity to capture features. Furthermore, the proposed model exhibits the highest GAN-train score in comparison to its competitors, which suggests that the proposed model demonstrates an advantage in terms of sample generation diversity and realism. It is noteworthy that AAE performs well on the GAN-train metric but poorly on the GAN-test metric, indicating that it learns the distribution of the original samples but has limited fitting ability. This results in a less accurate classifier when trained with real samples and tested with generated samples. The other models exhibit smaller differences in GAN-train and GAN-test metrics, but these are lower than the proposed model, which provides evidence for the superiority of the proposed model.

Fig. 11 illustrates the classification accuracies of the recently trained diagnostic models for each diagnostic task (Table 6) in Case 2. Based on this data, several conclusions can be drawn. Firstly, the average test accuracy of the DCGAN method is significantly lower at 76.55 %. This suggests that there is interference between the data distributions during the training process, which affects the model's ability to generalize. Secondly, the GAN methods, including AAE, Vanilla GAN and WGAN-GP, demonstrate superior performance compared to other models, with average accuracies of 85.97 %, 85.50 % and 89.43 % in Case 2, respectively. Ultimately, the proposed method demonstrated the most effective diagnostic performance across the majority of diagnostic tasks, with an average accuracy of 95.01 %. Moreover, the method demonstrated greater stability in the majority of cases in comparison to the other methods.

To assess the effectiveness of the proposed diagnostic framework, the generated signals comparing the real signals of damage type of OF1, IF1 and CF are illustrated in Fig. 11. Notably, there were no fault samples with a 1000 N load and an rpm of 40 Hz in the training set. Thus, DiffUCD framework can generate fault signals for unseen combinations of rpm and load.

Moreover, the frequency domain comparison between the generated signals and the real signals shows that the frequency spectrum of the generated signal and the real signal has a high degree of coincidence as shown in Fig. 12. For the characteristic frequency of bearing failures, the

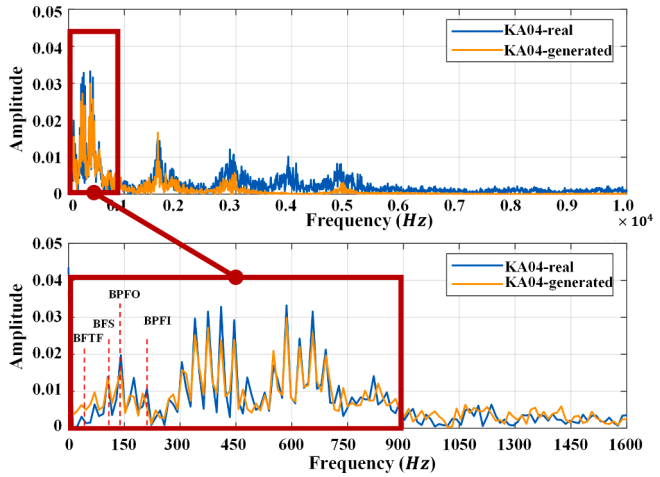


Fig. 13. Frequency domain comparison between the generated signals and the real signals in Case 2.

highest frequency is the ball pass frequency of inner race (BPFI), which can be calculated by the following formulation:

$$f_{BPFI} = \frac{nf_s}{2}\left\{1 + \frac{d}{D}cos\alpha\right\} \#(17)$$

where D, d, α, n and f_s are the pitch diameter, ball diameter, contact angle between the ball and the cage, number of rolling elements and rotating frequency of bearing. Since the highest rotation frequency in PU dataset is 25 Hz, the fault characteristic frequency is about 9.89 f_s, the highest characteristic fault frequency is 247.26 Hz. In the comparison of frequency domain in Fig. 13., the low frequency band covering 0 ~ 1200 Hz have a high degree of similarity between the unseen real signals and the generated signals.

As a conclusion, it can be found that the proposed method can generate signals of unseen working conditions, and they are very similar to the real signal of both time and frequency aspects. The effectiveness and practicability of the generative model is verified.

Furthermore, the effectiveness of generated signals is verified. Fig. 14. depicts the features of unseen working conditions with the dark blue color of the proposed method from different tasks. As shown in Fig. 14. (a) and (b), the clear decision boundary of each fault can be found to facilitate diagnosis, the generated signals process the similar feature distributions of the same fault types. However, as shown in panels (b) and (c), although the unseen condition is inaccessible, the learned features are domain-invariant and fault-related.

4.3. Hyperparameter sensitivity analysis

The performance of DiffUCD framework is strongly influenced by the hyperparameters set. Initial model architecture and parameters are set as Table 8. Adjustable parameters are involved in the construction and training of the proposed method and considering their impact on the model performance, parameter sensitivity analyses are performed on all case. In this part, hyperparameters of two aspects including the whole framework and the diffusion process are discussed.

7) Sensitivity analysis of DiffUCD framework.

Diffusion step t and UCFilter selection parameter k are two adjustable hyperparameters in the DiffUCD framework. As for the parameter of diffusion steps, the larger the value of diffusion steps the more detailed the signal generation process will be, as well as the larger the model training time and the computational resources it will consume.

In the training phase, comparative experiments with diffusion steps as {500,750,1000,1250,1500} are performed with the k value fixed as 0.7. As shown in Fig. 14 (a), it can be concluded that the performance of DiffUCD framework becomes better and more stable as t is bigger than

Table 8
Main architecture and parameters setting.

Main architecture and parameter	Value
Number of Condition Encoder layers	3
Number of Attention Encode layers	3
Number of Attention Decode layers	3
Diffusion Steps t	1200
β_1 in Reverse Process	0.001
β_T in Reverse Process	0.05
k in UCFilter Selection	0.7

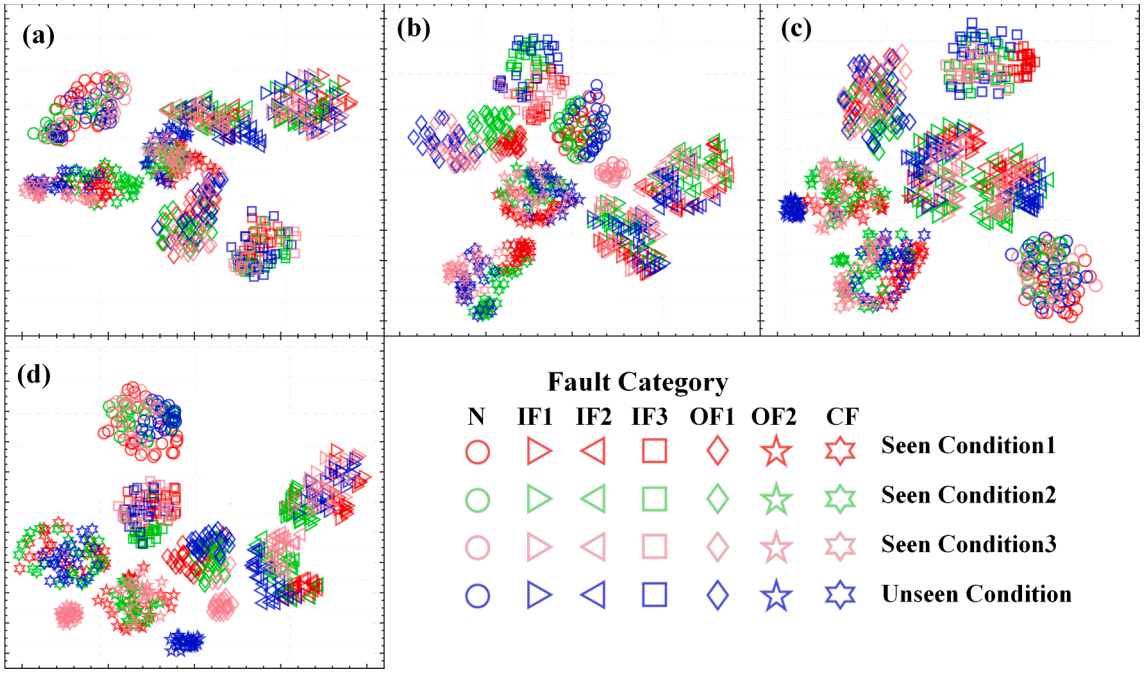


Fig. 14. Features distribution with generated signals under unseen working condition: (a) $p1$, (b) $p2$, (c) $p3$, and (d) $p4$.

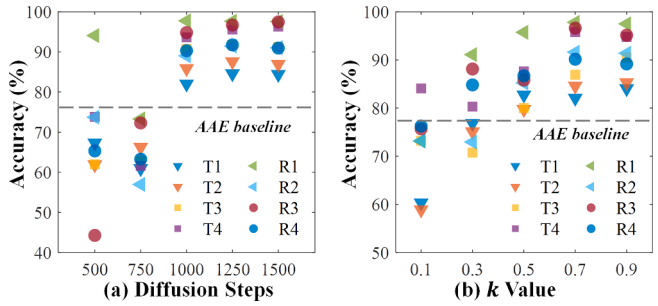


Fig. 15. Frequency domain comparison between the generated signals and the real signals in Case 2.

1000 steps. Regarding the performance of AAE method as the baseline, the diffusion steps should not be less than 1000. As t increased from 1000 to 1500, the performance of the model stabilizes and does not improve significantly.

k value represents the proportion of selected samples ($k = n/N$), where N is the total number of generated samples and n is the number of selected samples among them. Fig. 15 (b) illustrates the diagnostic accuracy for each case under different k values ranging from 0.1 to 0.9. The graph indicates that when k is below 0.5, meaning less than half of the total generated samples are selected, the performance is inferior to the baseline M1 method. This could be attributed to the reduced generation capability resulting from a smaller set of selected samples, causing offsets in the feature distribution from the real distribution. Conversely, if k is too large, the model's performance declines, likely due to the utilization of too many substandard generated signals for diagnostic model training. Hence, selecting a value of k around 0.7 is recommended.

8) Sensitivity analysis of denoising process.

The β_t in reverse diffusion process is computed using $1-\alpha_t$ as in Equ. (3), means the disturbance rate when denoising the generated sample. Diffusion model employs a multistep denoising process to gradually remove noise from a noisy sample. β_1 is the initial denoise value up to α_T while β_T is the max denoise value up to α_1. For the sensitivity analysis of denoise process, comparative experiments with variable β_1 set as {1e-06, 5e-06, 1e-05, 5e-05, 0.0001, 0.0005, 0.001, 0.005} and β_T set as {0.0001, 0.0005, 0.001,0.005,0.01,0.05,0.1,0.5}. As shown in Fig. 16., Among all 64 hyperparameter combinations between β_1 and β_T, the GAN-Train test accuracy is of clearly fluctuates considerably as both the parameter varies.

Whereas, the value of β_1 process more influence on the diagnostic accuracy. When the value of β_1 is larger than 5e-05 smaller than 0.0005 and β_T is larger than 0.001 and smaller than 0.005, it is easier to train a stable diffusion model.

In the structure of CGE-UNet, the condition encoder converts text conditional labels to embeddable vector and the embedded into diffusion process through the encode/decode attention block. The complexity of the diffusion model up to the depth of condition encoder and encode/decode attention block. Initially, as shown in Table 8, the depth of these two networks is set as 3, which means the number of the layers in condition encoder and the numbers of both encode/decode attention block. Hence, a parallel search for network layers is performed, where the depth is set from 2 to 10. The GAN-test accuracy is shown in Fig. 17, from which it is can be seen that the layers of condition encoder achieve acceptable accuracy as {3,7} layers and the numbers of encoder/decoder attention block are {3,4,9,10}. Taking the training cost into consideration, the combination of 3 layers of condition encoder and 3 or 4 layers of encoder/decoder attention block is the most suitable

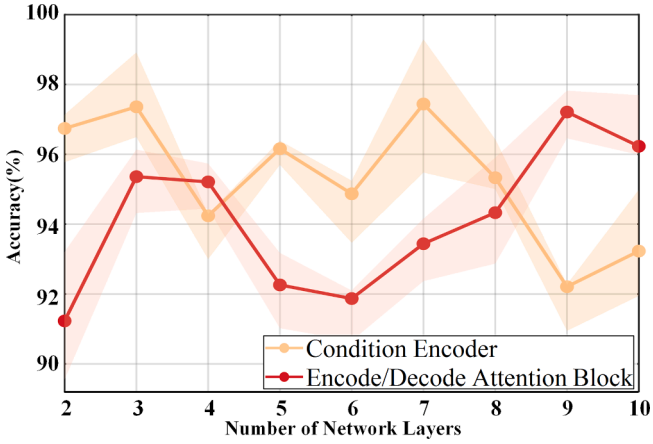


Fig. 17. The accuracy test under varies network layers.

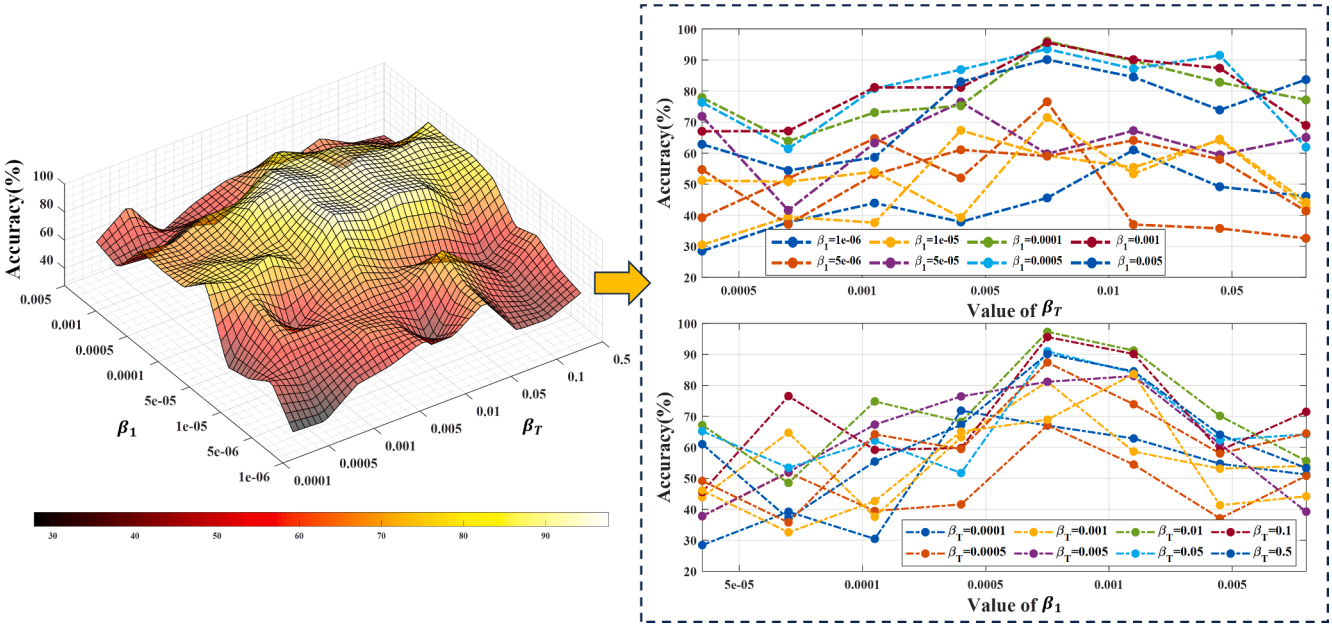


Fig. 16. Diagnosis accuracy obtained using different values of the hyperparameters β_1 and β_T.

hyperparameter in CGE-UNet structure.

Ultimately, in the aspect of proposed DiffUCD framework, it can be concluded that the diffusion steps, the k value in UCFilter, the reverse denoising weights and the CEG-UNet structures can be adjusted for the best generation state as discussed above.

5. Conclusion

In conclusion, the DiffUCD framework for diagnosing bearing faults under unseen working conditions for safety critical equipment is proposed. Leveraging a generative diffusion model and condition-guided information (CGI), this framework effectively embeds signal features under various working conditions, and the UCFilter method ensures signal quality using principles from k-means clustering. Diffusion model continuously learns the noise data added at each step through a forward diffusion process and then generates samples with embedded condition information for the unseen working condition diagnosis. Extensive fault diagnosis experiments are conducted on the PU and SDUST rolling bearing dataset. The proposed method is thoroughly compared with the state-of-the-art generative models in the realm of fault diagnosis. Two evaluation methods (MMD score and GAN-train/test) are utilized for generated signal evolutions. The experimental outcomes indicate the notable superiority of our model in terms of generating a wider array of high-quality samples for unseen working conditions than other methods. The main contribution of this article can be summarized as follows:

1) A thoughtfully designed working condition embedding method CGE-UNet is integrated within the DiffUCD framework based on attention mechanism, which effectively leveraging the working condition information into the diffusion process.
2) The comparison and feature visualization results with other state-of-the-art generative methods prove the superiority of DiffUCD for embedding unseen domain features and the flexible signal generative ability.
3) Through the hyperparameter analysis, the generation ability and the robustness of our model are further validated. Moreover, the best range of parameter settings are preliminarily given by designed experiments.

In essence, the proposed DiffUCD framework adeptly working condition shift challenges stemming from varying loads and rpms. Furthermore, the model's flexibility allows operators to customize adjustable thresholds, achieving a practical balance between performance and computing resources for real-world applications.

CRediT authorship contribution statement

Zhibin Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lefei Xu:** Visualization, Validation, Conceptualization. **Yuhao Zheng:** Resources, Conceptualization. **Jingsong Xie:** Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Tiantian Wang:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] E. Zio, Prognostics and Health Management (PHM): Where are we and where do we (need to) go in theory and practice, Reliability Engineering & System Safety 218 (2022).
[2] M. Kordestani, M. Saif, M.E. Orchard, R. Razavi-Far, K. Khorasani, Failure Prognosis and applications-a survey of recent literature, Ieee Transactions on Reliability 70 (2) (2021) 728–748.
[3] Z. Guo, J. Xie, T. Pan, T. Wang, Robust Bearing Fault Diagnostic Framework Based on Diffusion Model for Unknown Working Conditions. 8th International Conference on Condition Monitoring of Machinery in Non-Stationary Operations, 2024.
[4] L. Chen, Q. Li, C.Q. Shen, J. Zhu, D. Wang, M. Xia, Adversarial domain-invariant generalization: a generic domain-regressive framework for bearing fault diagnosis under unseen conditions, Ieee Transactions on Industrial Informatics 18 (3) (2022) 1790–1800.
[5] Z. Guo, T. Wang, J. Xie, J. Yang, Q. Peng, A deep transfer learning-based open scenario diagnostic framework for rail damage using ultrasound guided waves, Ieee Transactions on Instrumentation and Measurement 73 (2024) 1–17.
[6] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J.W. Vaughan, A theory of learning from different domains, Machine Learning 79 (1–2) (May 2010) 151–175.
[7] J. Berner, P. Grohs, A. Jentzen, Analysis of the Generalization Error: Empirical Risk Minimization over Deep Artificial Neural Networks Overcomes the Curse of Dimensionality in the Numerical Approximation of Black-Scholes Partial Differential Equations, Siam Journal on Mathematics of Data Science 2 (3) (2020) 631–657.
[8] X.J. Guo, L. Chen, C.Q. Shen, Hierarchical adaptive deep convolution neural network and its application to bearing fault diagnosis, Measurement 93 (Nov 2016) 490–502.
[9] Z.Y. Chen, W.H. Li, Multisensor Feature Fusion for Bearing Fault Diagnosis Using Sparse Autoencoder and Deep Belief Network, Ieee Transactions on Instrumentation and Measurement 66 (7) (Jul 2017) 1693–1702.
[10] X. Wang, C.Q. Shen, M. Xia, D. Wang, J. Zhu, Z.K. Zhu, Multi-scale deep intra-class transfer learning for bearing fault diagnosis, Reliability Engineering & System Safety 202 (Oct 2020).
[11] C.F. Hu, Y.X. Wang, J.W. Gu, Cross-domain intelligent fault classification of bearings based on tensor-aligned invariant subspace learning and two-dimensional convolutional neural networks, Knowledge-Based Systems 209 (Dec 2020).
[12] Y. Zhuo, Z.Q. Ge, Auxiliary Information-Guided Industrial Data Augmentation for Any-Shot Fault Learning and Diagnosis, Ieee Transactions on Industrial Informatics 17 (11) (Nov 2021) 7535–7545.
[13] X. Li, W. Zhang, Q. Ding, J.Q. Sun, Intelligent rotating machinery fault diagnosis based on deep learning using data augmentation, Journal of Intelligent Manufacturing 31 (2) (Feb 2020) 433–452.
[14] S.N. Tang, S.Q. Yuan, Y. Zhu, Data Preprocessing Techniques in Convolutional Neural Network Based on Fault Diagnosis Towards Rotating Machinery, Ieee Access 8 (2020) 149487–149496.
[15] W. Zhang, X. Li, H. Ma, Z. Luo, X. Li, Open-Set Domain Adaptation in Machinery Fault Diagnostics Using Instance-Level Weighted Adversarial Learning, Ieee Transactions on Industrial Informatics 17 (11) (Nov 2021) 7445–7455.
[16] J. Luo, J.Y. Huang, H.M. Li, A case study of conditional deep convolutional generative adversarial networks in machine fault diagnosis, Journal of Intelligent Manufacturing 32 (2) (Feb 2021) 407–425.
[17] Y. Xia, C.Q. Shen, D. Wang, Y.J. Shen, W.G. Huang, Z.K. Zhu, Moment matching-based intraclass multisource domain adaptation network for bearing fault diagnosis, Mechanical Systems and Signal Processing 168 (Apr 2022).
[18] K.W. Wang, X. Zhang, Q.S. Hao, Y. Wang, Y. Shen, Application of improved least-square generative adversarial networks for rail crack detection by AE technique, Neurocomputing 332 (Mar 2019) 236–248.
[19] H. Y. Wang, P. Li, X. Lang, D. P. Tao, J. Ma, and X. Li, "FTGAN: A Novel GAN-Based Data Augmentation Method Coupled Time-Frequency Domain for Imbalanced Bearing Fault Diagnosis," *Ieee Transactions on Instrumentation and Measurement,* vol. 72, 2023.
[20] J. Gui, Z.A. Sun, Y.G. Wen, D.C. Tao, J.P. Ye, A Review on Generative Adversarial Networks: Algorithms, Theory, and Applications, Ieee Transactions on Knowledge and Data Engineering 35 (4) (Apr 2023) 3313–3332.
[21] Z.L. Huang, Z.H. Lei, G.R. Wen, X. Huang, H.X. Zhou, R.Q. Yan, X.F. Chen, A Multisource Dense Adaptation Adversarial Network for Fault Diagnosis of Machinery, Ieee Transactions on Industrial Electronics 69 (6) (Jun 2022) 6298–6307.
[22] K. Rombach, G. Michau, O. Fink, Controlled generation of unseen faults for Partial and Open-Partial domain adaptation, Reliability Engineering & System Safety 230 (Feb 2023).
[23] C.W. Bang, C. Chun, Effective Zero-Shot Multi-Speaker Text-to-Speech Technique Using Information Perturbation and a Speaker Encoder, Sensors 23 (23) (Dec 2023).
[24] S.X. Jia, J.R. Wang, B.K. Han, G.W. Zhang, X.Y. Wang, J.T. He, A Novel Transfer Learning Method for Fault Diagnosis Using Maximum Classifier Discrepancy With Marginal Probability Distribution Adaptation, Ieee Access 8 (2020) 71475–71485.
[25] Z. Guo, J. Xie, T. Pan, T. Wang, DiffPhysiNet: A Bearing Diagnostic Framework Based on Physics-Driven Diffusion Network for Unseen Working Conditions, PHM Society European Conference (2024).
[26] X.Y. Yang, T.Y. Ye, X.F. Yuan, W.J. Zhu, X.X. Mei, F.Y. Zhou, A Novel Data Augmentation Method Based on Denoising Diffusion Probabilistic Model for Fault

Diagnosis Under Imbalanced Data, Ieee Transactions on Industrial Informatics 20 (5) (May 2024) 7820–7831.
[27] B. Yang, Y.G. Lei, F. Jia, N.P. Li, Z.J. Du, A Polynomial Kernel Induced Distance Metric to Improve Deep Transfer Learning for Fault Diagnosis of Machines, Ieee Transactions on Industrial Electronics 67 (11) (Nov 2020) 9747–9757.
[28] Z. An, S.M. Li, J.R. Wang, X.X. Jiang, A novel bearing intelligent fault diagnosis framework under time-varying working conditions using recurrent neural network, ISA Transactions 100 (Oct 2020) 155–170.
[29] C. Lessmeier, J.K. Kimotho, D. Zimmer, W. Sextro, Condition Monitoring of Bearing Damage in Electromechanical Drive Systems by Using Motor Current Signals of Electric Motors: A Benchmark Data Set for Data-Driven Classification, PHM Society European Conference (2016).
[30] Z. Q. Pu, D. Cabrera, C. A. Li, and J. V. de Oliveira, "VGAN: Generalizing MSE GAN and WGAN-GP for Robot Fault Diagnosis," *Ieee Intelligent Systems,* vol. 37, no. 3, pp. 65-75, May-Jun, 2022.
[31] W. Fang, F.H. Zhang, V.S. Sheng, Y.W. Ding, A Method for Improving CNN-Based Image Recognition Using DCGAN, Cmc-Computers Materials & Continua 57 (1) (Oct 2018) 167–178.
[32] Q. M. Jin, R. H. Lin, F. C. Yang, "E-WACGAN: Enhanced Generative Model of Signaling Data Based on WGAN-GP and ACGAN," *Ieee Systems Journal,* vol. 14, no. 3, pp. 3289-3300, Sept, 2020.
[33] F. Ye, and A. G. Bors, "Lifelong Generative Adversarial Autoencoder," *Ieee Transactions on Neural Networks and Learning Systems*, 2023 Jul, 2023.
[34] L. van der Maaten, G. Hinton, Visualizing Data using t-SNE, J. Mach. Learn. Res. 9 (2008) 2579–2605.